

PAPER_202: UQFF Reionization, BBN, Recombination, and Cosmic Dawn Physics

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$$F_U(r, t) = \sum_{i=1}^4 U_{gi} + U_m + U_A - U_{b_i}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, [SSq] = 0.57$$

$$U_{b_i}(r) = \kappa \cdot [SSq] \cdot \frac{GM}{r^2}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, [SSq] = 0.57, \beta_i = 0.61$$

Abstract

Key UQFF calibrated constants: $\kappa = 5.0\text{e-}4 \text{ day}^{-1}$; $[SSq] = 5.7\text{e-}1$; $H_{SCm} \approx 9.9\text{e-}1$; $U_{UA} \approx 1.0\text{e-}4$; $k_{\eta} = 1.0\text{e-}113$; $\beta_i \approx 6.0\text{e-}1$; $G = 6.674\text{e-}11 \text{ N}\cdot\text{m}^2/\text{kg}^2$

This paper presents the UQFF buoyancy formalism applied to cosmic dawn and reionization physics: baryon-to-photon ratio (?), deuterium bottleneck during Big Bang Nucleosynthesis (BBN), CMB angular power spectrum, recombination optical depth, ionization fraction evolution, HII bubble growth rate, and Jeans mass/wavelength for fragmentation. These collectively span the redshift range $z \sim 1100$ (recombination) through $z \sim 5$ (end of reionization). Each F_{UBii} and U_m variant is rigorously derived from observational anchor equations.

UQFF Discovery: Novel application of UQFF calibration constants ($\kappa = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Baryon-to-Photon Ratio (?)

$$\eta = n_b/n_{\gamma} = 6.08 \times 10^{-10} \quad (\text{Planck 2018} + \text{BBN fit})$$

$$F_{UBii, \eta} = F_{rel} \times (\eta = n_b/n_{\gamma} / E_{LEP}) \times Q_{wave} \times [\text{Neff modification}]$$

$$U_{m, \eta}(t) = \mu(\eta_{vac}) \cdot (1 - e^{-\eta t}) \cdot \eta$$

Physical origin:

$$n_{\gamma} = 410 \text{ cm}^{-3} \quad (\text{CMB photon number density today})$$

Fit anchor: Primordial D, ^3He , ^4He , ^7Li abundances
 ? determines all light element yields (standard BBN)
 UQFF role: ? sets vacuum energy scale through $\gamma_b/\gamma_?$ at nucleosynthesis

2. Deuterium Bottleneck (BBN Timing)

Deuterium bottleneck timescale:

$$t_D \sim v(3/(32pG\gamma_{\text{rad}})) \sim 180 \text{ s at } T \sim 0.1 \text{ MeV}$$

$$\gamma_{\text{rad}} = p^2 k T^4 / (30 h^3 c^5) \cdot g_* \quad (\text{effective DOF } g_* \sim 10 \text{ at D-formation onset})$$

Sequence:

$T \sim 1 \text{ MeV}$: neutron freeze-out ($n/p \sim 1/7$)

$T \sim 0.1 \text{ MeV}$ ($t \sim 180 \text{ s}$): D photodissociation ends ? ^4He chains proceed

Yield: $Y_P \sim 0.247$ (mass fraction ^4He)

$$F_{\text{UBii,deb}} = -F_{\text{rel}} \times (t_D / E_{\text{LEP}}) \times Q_{\text{wave}} \times [g_*(T) \text{ variation}]$$

$$U_{\text{m,deb}}(T) = \mu(\gamma_{\text{vac}}) \cdot (1 - e^{-\gamma t}) \cdot [\text{Weak freeze at } T \sim 1 \text{ MeV; D photodissociation below}]$$

UQFF calibration: ? calibrated to UQFF $k_?$ parameter through BBN freeze-out

$k_? \sim 10^{11.3}$ (small-scale vacuum fluctuation coupling)

$\gamma_{k_?}$ (CIA cross-section refit) $\sim 7.25 \times 10^8$ (fractional shift from H2 CIA data)

3. CMB Angular Power Spectrum

$$C_l = (2/p) \cdot \gamma k^2 dk \cdot P(k) \cdot |\gamma_l^T(k)|^2$$

$P(k) \propto k^{n_s-4}$ (primordial, $n_s \sim 0.965$ Planck 2018)

$\gamma_l^T(k)$ = transfer function:

- Large scales ($l < 100$): Sachs-Wolfe plateau $C_l \sim \text{constant}$
- Acoustic peaks ($l = 220, 540, 800 \dots$): baryon-photon oscillations
- Damping tail ($l > 1000$): Silk damping

$$F_{\text{UBii,cmb}} = F_{\text{rel}} \times (C_l \cdot l(l+1)/(2p) / E_{\text{LEP}}) \times Q_{\text{wave}}$$

$$U_{\text{m,cmb}}(k) = \mu(\gamma_{\text{vac}}) \cdot (1 - e^{-\gamma t}) \cdot \text{Transfer } \gamma_l^T: \text{Sachs-Wolfe + acoustic oscillations}$$

Calibration anchor:

$$D_l = l(l+1)C_l/(2p) \times T_0^2 \propto \text{peak at } l \sim 220 \text{ matches } 0_{\text{tot}} = 1$$

UQFF role: vacuum energy $\gamma c^2/3$ term in g_{UQFF} sets acoustic horizon

4. Recombination Optical Depth

$$\tau(z) = \int_z^{\infty} n_e(z') \cdot s_T \cdot c \cdot |dt/dz'| dz'$$

$$s_T = 6.652 \times 10^{-29} \text{ m}^2 \quad (\text{Thomson cross-section})$$

$$n_e(z) = x_e(z) \cdot n_H(z) = x_e(z) \cdot n_{b,0} \cdot (1+z)^3$$

Recombination: $z_{\text{rec}} \sim 1100$ $\tau(z_{\text{rec}}) \sim 1$ (photon decoupling)

Reionization: $z_{\text{re}} \sim 7.7$ optical depth $\tau_{\text{re}} \sim 0.054$ (Planck 2018)

$$F_{\text{UBii, recomb}} = -F_{\text{rel}} \times (\tau(z) / E_{\text{LEP}}) \times Q_{\text{wave}}$$

$$U_{\text{m, recomb}}(z) = \mu(\tau_{\text{vac}}) \cdot (1 - e^{-\tau}) \cdot n_e \cdot s_T \cdot c \cdot dt$$

Physical context:

After z_{rec} , photons decouple and stream freely (CMB)

Surface of last scattering has $T/T \sim 10^5$ (measured by Planck)

5. Reionization: Ionization Fraction Evolution

$$dx_e/dt = \tau_{\text{eff}} \cdot e_{\text{esc}} \cdot f_* - a_B \cdot n_e^2 \cdot C$$

where:

$$\tau_{\text{eff}} = (1+z)^3 \cdot n_b \cdot \tau_{\text{eff}} \quad (\text{ionizing photon rate})$$

$$e_{\text{esc}} \sim 0.1-0.3 \quad (\text{photon escape fraction from galaxies})$$

$$f_* \sim 0.05-0.2 \quad (\text{star formation efficiency in halos})$$

$$a_B = 2.6 \times 10^{-13} \text{ cm}^3/\text{s} \quad (\text{recombination coefficient, case B, } T \sim 10^4 \text{ K})$$

$$C = \tau_{\text{H}}^2 / n_{\text{H}}^2 \quad (\text{clumping factor, } C \sim 3)$$

$$F_{\text{UBii, ion}} = F_{\text{rel}} \times (\tau_{\text{eff}} \cdot e_{\text{esc}} \cdot f_* / E_{\text{LEP}}) \times Q_{\text{wave}} \times [\text{recombination subtracted}]$$

$$U_{\text{m, ion}}(t) = \mu(\tau_{\text{vac}}) \cdot (1 - e^{-\tau}) \cdot a_B \cdot n_e^2 \cdot C \cdot dt$$

Reionization history:

$z \sim 20-30$: first stars ionize H around them (PopIII)

$z \sim 7-9$: bulk reionization, x_e rises from ~ 0.1 to ~ 1

$z \sim 5-6$: completion (Gunn-Peterson trough in quasar spectra)

6. HII Bubble Growth During Reionization

$$dN_b/dt = \tau_{\text{eff}} \cdot e_{\text{esc}} - a_B \cdot n_H \cdot (4/3 \pi R_b^3)$$

Simplified overlap model (Stromgren sphere analog):

$$R_b(t) = (3 \tau_{\text{eff}} \cdot t / (4 \pi \cdot n_H))^{1/3} \quad (\text{linear ionization front growth})$$

Γ_{γ} = rate of ionizing photons emitted (from galaxy SFR)
 n_H = neutral H density (z-dependent)

$$F_{UBii,bub} = F_{rel} \times (R_b(t) / E_{LEP}) \times Q_{wave} \times (\Gamma_{\gamma} \cdot x_e)$$

$$U_{m,bub}(t) = \mu(\Gamma_{vac}) \cdot (1 - e^{-\Gamma_{\gamma} t}) \cdot [\text{Expand for moving ionization front}]$$

Physical context:

Bubble merger $\rightarrow$ percolation at $x_{HI} < 0.1 \rightarrow$ full reionization

UQFF buoyancy at bubble edge acts as expansion driver ($F_{UBii,bub} > 0$)

7. Jeans Length and Mass (Fragmentation)

Jeans length:

$$\lambda_J = p^{1/2} \cdot c_s / (G \cdot \rho)^{1/2}$$

Jeans mass:

$$M_J = (p/6) \cdot \rho \cdot \lambda_J^3 = (5k_{BT}/(G\mu_H))^{3/2} \cdot (3/(4p))^{1/2}$$

Stability condition: $\rho > \rho_J \rightarrow$ gravitational collapse

Dispersion relation: $\omega^2 = c_s^2 k^2 - 4\pi G \rho$

Unstable: $\omega^2 < 0 \rightarrow k < k_J = 2\pi/\lambda_J$

$$F_{UBii,jeans} = -F_{rel} \times (\lambda_J / E_{LEP}) \times Q_{wave} \times (\text{collapse onset time } t \sim 1/v(G\rho))$$

$$U_{m,jeans}(T) = \mu(\Gamma_{vac}) \cdot (1 - e^{-\Gamma_{\gamma} t}) \cdot [\text{Perturb: } \omega^2 = c_s^2 k^2 - 4\pi G \rho]$$

Physical context:

First (PopIII) stars: $T \sim 200$ K, cloud $M \sim 100\text{--}1000 M_{\odot} \gg M_J$

Present-day GMCs: $T \sim 10\text{--}30$ K $\rightarrow M_J \sim 1\text{--}10 M_{\odot}$

UQFF buoyancy at Jeans scale: $F_{UBii,jeans}$ acts as tidal disruption force

8. Alfvén Wave and Turbulent Cascade

Alfvén wave velocity:

$$v_A = B/v(4\pi\rho) \quad (\text{for } B \text{ in Gauss, } \rho \text{ in g/cm}^3)$$

Anisotropic cascade (Goldreich-Sridhar):

$$k_{\perp}/k_{\parallel} \sim (k_{\perp} \cdot l_A)^{1/3} \quad (\text{scale-dependent anisotropy})$$

$$F_{UBii,alf} = F_{rel} \times (v_A / E_{LEP}) \times Q_{wave} \times (B \cdot v_A) \times e^{-t/t_{eddy}}$$

Turbulent energy cascade (Kolmogorov in ISM):

$$e = v_l^3/l = \text{constant} \quad (\text{energy transfer rate per unit mass})$$

$v_l \propto l^{1/3}$ (velocity-scale relation)
Power spectrum: $E(k) \propto k^{-5/3}$

$$F_{UBii,turb} = F_{rel} \times (e^{2/3} \cdot l^{-2/3} / E_{LEP}) \times Q_{wave} \times e^{-kl/k_J}$$

$$U_{m,alf(B)} = \mu(\tau_{vac}) \cdot (1 - e^{-\tau}) \cdot v_A(B)$$

$$U_{m,turb(l)} = \mu(\tau_{vac}) \cdot (1 - e^{-\tau}) \cdot (\tau \cdot v_l^3 / l)$$

9. Ionization Parameter and Feedback Coupling

Ionization parameter:

$$U = Q_H / (4\pi \cdot r^2 \cdot n_H \cdot c)$$

where:

Q_H = number of ionizing photons per second (from AGN or OB stars)

r = distance from ionizing source

$\log U \sim -2$ to -3 for NLR, -1 to 0 for BLR

AGN feedback coupling efficiency:

$$e_f = E_{kin} / (\tau_{acc} \cdot c^2) \sim 0.05 - 0.1$$

$$E_{kin} = (1/2) \tau_{out} \cdot v^2_{out}$$

Only $e_f \times 10^{-5}$ per $M_\odot$ needed to match M_{BH-s}

$$F_{UBii,upar} = F_{rel} \times (U \cdot n_H \cdot c / E_{LEP}) \times Q_{wave} \times (Q_H / r^2)$$

$$F_{UBii,coup} = F_{rel} \times (e_f \cdot \tau_{acc} \cdot c^2 / E_{LEP}) \times Q_{wave} \times [0.05 - 0.1]$$

$$U_{m,upar(r)} = \mu(\tau_{vac}) \cdot (1 - e^{-\tau}) \cdot (Q_H / (4\pi r^2 n_H c))$$

$$U_{m,coup(\tau)} = \mu(\tau_{vac}) \cdot (1 - e^{-\tau}) \cdot e_f \text{ (couple fraction to kinetic energy)}$$

10. Cosmological Timeline Summary

Epoch	Redshift	UQFF Process	F_{UBii} Variant
Inflation	$z \sim 10^{27}$	Inflaton field $V(\phi)$	$F_{UBii,inf}$
Planck epoch	$z \sim 10^{32}$	LQC bounce	$F_{UBii,bounc}$
Neutron freeze-out	$z \sim 10^{10}$ ($T \sim 1$ MeV)	Weak interactions	$F_{UBii,eta}$
BBN	$z \sim 10^9$ ($t \sim 180$ s)	D bottleneck	$F_{UBii,deb}$
Recombination	$z \sim 1100$	Photon decoupling	$F_{UBii,recomb}$
Cosmic dawn	$z \sim 20-30$	First stars, first HII	$F_{UBii,ion}$
Reionization	$z \sim 6-9$	HII bubble percolation	$F_{UBii,bub}$

Epoch	Redshift	UQFF Process	F_UBii Variant
Present	$z \sim 0$	Structure + feedback	All F_UBii

11. References

- grok_share_7514fe.txt lines 6070–6090 (BB_C_Equations items 1310–1350)
- grok_share_7514fe.txt lines 6300–6500 (full reionization/BBN/CMB catalogue)
- PAPER_199: F_UBii Taxonomy Part 2 (Cosmological)
- PAPER_200: Um Universal Magnetism Catalogue
- Planck 2018 Collaboration: CMB and reionization parameters